Economic Complexity Extra Homework II. - Kolos Ágoston Mohácsi, GG9JD0

Data source: My goal was to replicate the methodology outlined in Hidalgo et al.'s (2007) product space, with the constraints of computation and data accessibility. For these purposes the United Nations (n. d.) COMTRADE database is used. In this database I collected HS2 level aggregated export volumes (in USD) data for every possible nation, for 2019 and 2024. Our export data contains volumes for 97 types of goods, for 176 nations for two years. From these data sources I calculated the RCA matrix, proximities and product spaces using Hidalgo et al.'s (2007) methodology.

Filtering: Following Hidalgo et al.'s (2007) work I restricted the product space network, by filtering the proximities by the network edges having one of the following two conditions. 1.) The edge proximity value is at least 35% or 2.) The edge is part of the largest spanning tree of the original network. This means that our network remains connected, while pruning as many weak proximities as possible.

Visualizing the networks:

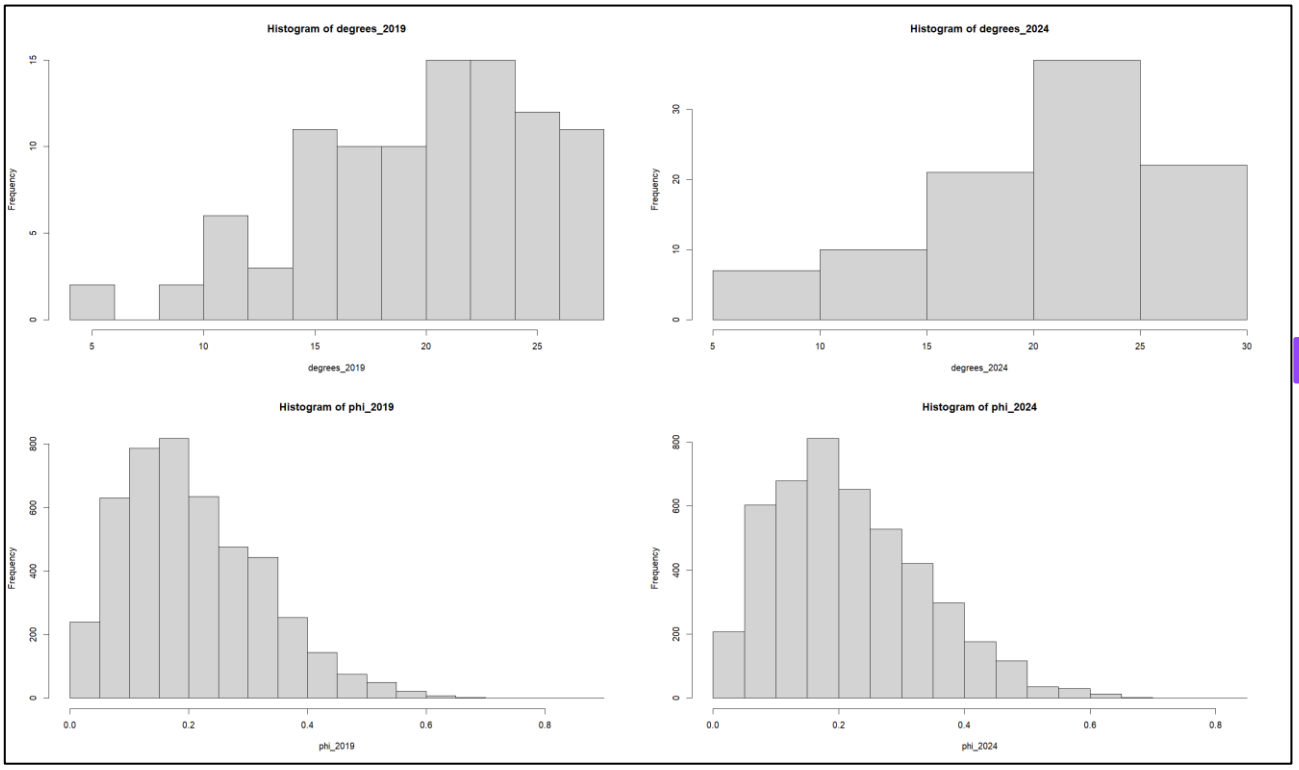


1. Image: Network plots of the 2019 & 2024 product spaces.

Source: United Nations (n. d.)

Looking at the 1. Image, the HS2 level product spaces are promising. The biggest similarity of with the key characteristics of Hidalgo et al.'s (2007) product spaces is the presence of one, large, strongly clustered core. This property is present in both year's networks.

Descriptive Network properties:

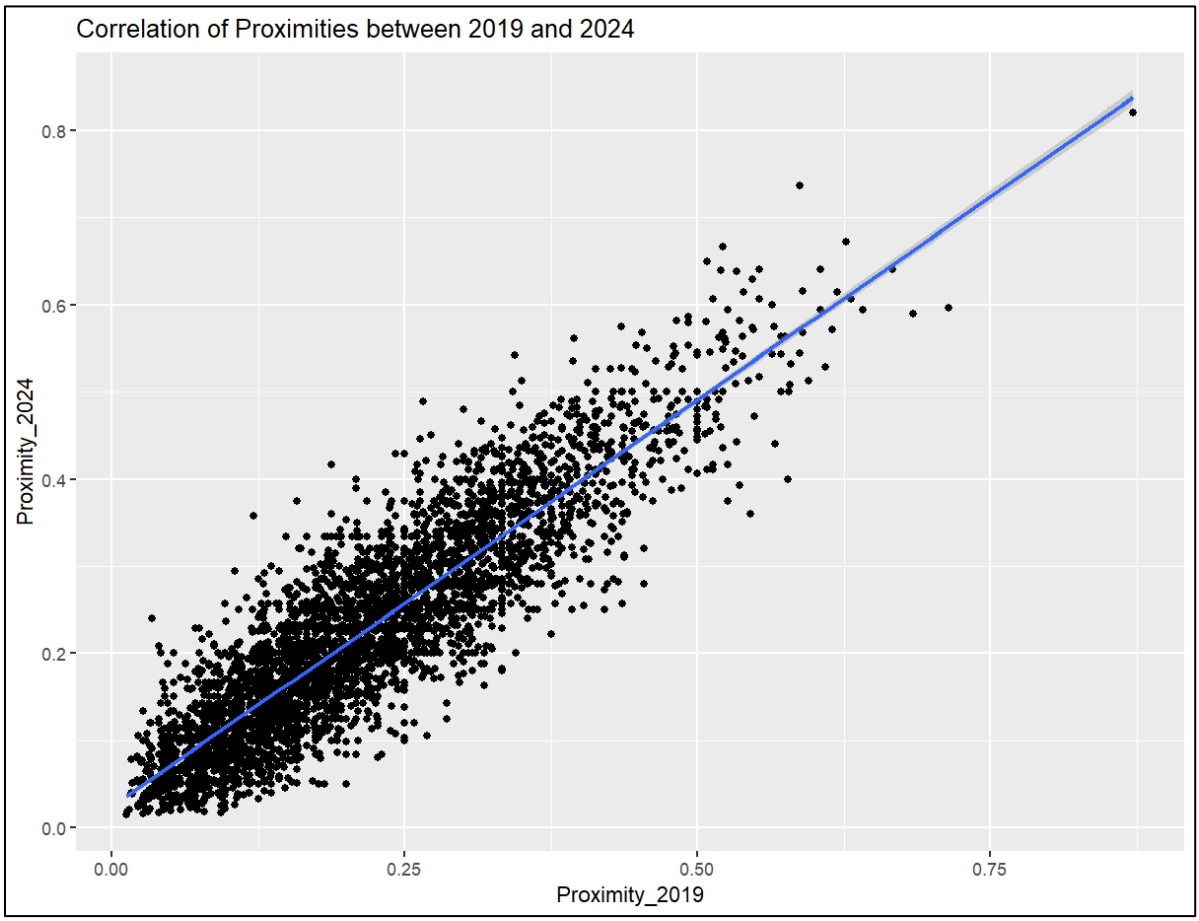


2. Image: Histogram of node degrees and edge proximities of 2019 and 2024.

Source: Based on United Nations (n. d.)

Analyzing degree and edgeweight distributions of the unfiltered product space network the 2. Image revealed several interesting aspects. The most noticeable is for the 2019 and 2024 proximity strength distribution as both have a very similar shape, being a right skewed distribution with the mode being around 20%. The proximity weighed degree histogram of the nodes have are less similar but their shapes also have parallels, notably the mode being 20-25 degrees with a left skewed distribution.

Correlation analysis of proximities:



3. Image: Scatterplot of the 2019 & 2024 Proximities.

Source: Based on United Nations (n. d.)

I joined the two networks non-filtered product space networks edgelists to compare the relation between the two years strength of proximities. The 3. Image reveals a very strong positive relation between the two years proximity values. Taking into the account the Pearson correlation coefficient (which value was 89,84%), also supports the strong nature of the positive relation. This might appear somewhat surprising as both the Covid, and Russo-Ukrainian war severely reoriented nation's roles in global supply chains – thus influencing RCA's - but the strong correlation implies that the product space networks relations are robust to such reorientations.

Bibliography

Hidalgo, C. A., Klinger, B., Barabási, A.-L. & Hausmann, R. (2007). The product space conditions the development of nations. Science, 317(5837), 482-487. 10.1126/science.1144581

United Nations (n. d.) *Trade Data.* [Database]. UN Comtrade database. Accessible: https://comtradeplus.un.org/TradeFlow Accessed: 2026.10.12.

AI Usage Statement

Tool used: *Claude (Anthropic), conversational interface*

What I used AI for:

1. Graph visualization troubleshooting: I asked Claude for help with dispersing a large igraph network plot that was collapsing into an unreadable blob. Claude explained the rescale = TRUE default behavior and suggested layout parameter tuning (graphopt charge, FR iterations, DRL).

2. Hidalgo product space methodology: I asked Claude to walk through the canonical pipeline for constructing Hidalgo-style product space visualizations — MST backbone extraction, proximity threshold filtering, and the visual encoding conventions (node size, sector color, country overlay).

3. igraph function troubleshooting: The `area` parameter in `layout_with_fr()` is defunct in recent igraph versions. Claude identified the issue and suggested workarounds (post-hoc coordinate scaling with rescale = FALSE, graphopt, DRL).

4. Basic igraph syntax: I asked how to extract node degrees (`degree()`, `strength()`).

What is my own contribution:

- The proximity matrix construction and data pipeline are my own work.

- The MST backbone + threshold graph was built by me following the methodology discussion.

- The choice of layout parameters and visual tuning (node sizing, edge transparency, threshold calibration) were iteratively adjusted by me based on the output.

- All R code was written and executed by me; Claude provided code snippets as suggestions which I adapted.